INSTRUMENT MANUAL

# Polymorph

Four-track synthesizer, drum machine and step sequencer
for live techno and experimental music.

POLYMORPH v0.33

**READY**

4 TRACKS · SYNTH · DRUMS · FX




VERSION 0.33

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**HOW THIS MANUAL IS WRITTEN**

Control names are printed like **CUTOFF**, keys like `Esc`, and messages on the amber display like `POLYMORPH READY`. Nothing in Polymorph is typed: every value is a knob, a button, a list, or a note you play.

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# 1 Welcome to Polymorph

Polymorph is a four-track synthesizer and drum machine with a step sequencer, built to be played live from a computer and a pair of small MIDI controllers.

Each of the four tracks is either a **SYNTH** track, with its own complete analog-style voice, or a **DRUM** track, with a kit of six synthesized drums. A sequencer of up to 64 steps per track drives them, with parameter locks, conditional trigs, microtiming, accent and slide. A tape echo, a reverb, a sidechain duck and a rumble generator finish the sound. Sixteen patterns can be switched in time with the bar.

## THE THREE PAGES

| PAGE | WHAT YOU DO THERE |
|---|---|
| **SYNTH** | Shape the sound of the selected track: oscillators, filter, contours, analog character, or the drum kit. Also the LFO, the mod matrix, the arpeggiator and RANDOM. |
| **SEQ** | Program and record patterns: tracks, steps, the step editor, the pattern bank and the files. |
| **FX** | Tape echo, reverb, rumble, sidechain duck, groove settings, audio setup and MIDI. |

The top strip (display, transport, clock) and the mixer strip are always visible above the page, and the bottom bar (PANIC, audio, output) below it.




*The SYNTH page with a SYNTH track selected. Top: display, transport and clock. Below: the mixer strip, then the page itself.*

## THE SOUND IN ONE PICTURE

**OSC 1** + **OSC 2** + **SUB** + **NOISE** → **DRIVE** → **LADDER FILTER** → **LOUDNESS CONTOUR** → **TRACK LEVEL** → **ECHO · REVERB** → **MASTER** → **LIMITER**

DRUM tracks follow the same path from the track level on: each drum has its own level, pan and sends, the kit goes through its own filter, and everything meets at Master. A limiter that looks 1.5 ms ahead keeps the output just under full scale without colouring the sound.

# 2 Getting started

## INSTALLING

Polymorph is a standalone Windows program. Copy the `Polymorph` folder anywhere on the computer (for example `C:\Polymorph`) and run **Polymorph.exe**. Nothing else needs to be installed. The first time, Windows may say "Windows protected your PC": choose **More info**, then **Run anyway**.

A small notice, `Preparing engine…`, appears first while the sound engine is made ready, so that the very first note you play is as quick as any other. It takes a second or two; the **first start on a new computer takes longer**, because the engine is compiled for that computer once and kept.

## SET WINDOWS TO 48 KHZ

Polymorph works at 48000 Hz. In Windows: **Settings › System › Sound**, click your output device, and set **Format** to **48000 Hz** (16 or 24 bit). Then close and reopen Polymorph. If no output is at 48 kHz, the display says `SET WINDOWS TO 48 KHZ` at start-up.

## FIRST SOUND

1. Connect headphones or speakers, and your MIDI controllers, **before** opening the program.
2. Choose **OUT** in the bottom bar. The list opens upward and shows every output with its full name. An output printed faintly cannot be used as it is: click it and the display says why (usually `SET THIS OUTPUT TO 48 KHZ IN WINDOWS SOUND SETTINGS`).
3. Keep **MASTER** at 0.50 and your headphone volume low the first time.
4. Press **AUDIO ON**. The display says `POLYMORPH READY`. Nothing sounds until you play.
5. Play the computer keys A S D… or your pad controller (chapter 4).
6. Press **AUDIO OFF** when you finish, or just close the window. If something is not saved, Polymorph asks first.

If the output you used last time is not connected, the display says `SAVED OUTPUT MISSING` and the status line names the one in use instead.

## LATENCY: HOW SOON A PAD IS HEARD

Three settings on the FX page, **SETUP**, decide the time from hitting a pad to hearing it. They can only be changed while audio is off, and are remembered.

| SETTING | CHOICES | WHAT IT DOES |
|---|---|---|
| **SOUND SYSTEM** | Windows WASAPI, MME | Which part of Windows plays the audio. WASAPI answers far sooner. The OUT list shows the outputs of the chosen system. |
| **BLOCK** | 256, 512, 1024 samples | The audio is worked out in blocks; what you play is heard from the next one. Smaller answers sooner and costs more processor. |
| **CARD BUFFER** | LOW, HIGH | How much the sound card keeps in hand. LOW is quicker; HIGH is safer on a busy computer. |

Typical figures, measured on a laptop's built-in audio with a full four-track scene playing (your headphones or interface add their own):

| SOUND SYSTEM | BLOCK | CARD BUFFER | PAD TO SOUND |
|---|---|---|---|
| **MME** | 1024 | HIGH | about 230 ms |

| | | | |
|---|---|---|---|
| **MME** | 512 | LOW | about 110 ms |
| **WASAPI** | 512 | HIGH | about 100 ms |
| **WASAPI** | 1024 | LOW | about 36 ms |
| **WASAPI** | **512** | **LOW** | **about 30 ms (the default)** |
| **WASAPI** | 256 | LOW | about 27 ms, but much more processor |

> **RECOMMENDED**
>
> **WASAPI, 512, LOW.** If the status line ever says `MS OF AUDIO LOST`, or the display's HEALTH shows LOST above 0, choose Block 1024, Card buffer HIGH, or the **LIGHT LAPTOP** button (chapter 15).

## SETTINGS TRAVEL WITH THE FOLDER

The output, sound system, block, card buffer, full screen, LIGHT LAPTOP and the MIDI devices are kept in `user\settings.json` **per computer**. Carry the folder to another computer and it starts there with that computer's own settings; yours are still there when the folder comes back. What each MIDI device does (its role and its links) is kept by the device's name and travels with the folder.

# 3 The panel

A black panel between walnut cheeks: no scrolling, nothing to type. Every size is drawn sharp, at any Windows scaling.

## THE TOP STRIP

| PART | WHAT IT IS |
|---|---|
| **Display** | Amber, four lines (below). |
| **PAGE** | SYNTH, SEQ, FX. |
| **TRANSPORT** | PLAY, STOP, RESET, **REC** and **STEP REC** (steps can only be changed while one of them is lit, chapter 10). |
| **MIDI** | MAP: link hardware controls to the panel (chapter 13). |
| **OCTAVE** | OCT − / OCT + (also the `Z` and `X` keys). |
| **TEMPO · SWING · CLOCK** | The tempo knob, the global swing (chapter 11), INT or EXT clock. |

## THE DISPLAY

| LINE | SHOWS |
|---|---|
| **Top, small** | The page on the left. On the right **HEALTH**: `PEAK 43% · LOST 0 MS · DROPS 0` while audio is on (chapter 15). |
| **Large** | The control you are touching and its value, for two seconds; the same when a MIDI controller moves one. Otherwise the state, or short messages: `KEYS → T2`, `PATTERN 3`, `MUTE 2 ON`, `SAVED: my groove`. |
| **Status** | `PTN 01>03 T1 S05 2.3.1`: pattern playing, `>` the one waiting, `*` unsaved changes, track and step selected, bar.beat.sixteenth. On the right the tempo and INT or EXT. |
| **Bottom, small** | Muted and soloed tracks, the octave shift, the MIDI port. For a step with a condition: its rule in words. |

## THE MIXER STRIP

Always visible under the top strip: T1–T4 with M, S, Level, DUCK and a meter each; the LIVE channel; the MOD wheel; Master with two meters; **FX ON** and, under it, **MONO**. Chapter 7 explains it.

## KNOBS, BUTTONS AND LISTS

| TO | DO |
|---|---|
| **Turn a knob** | Drag up or down on it. |
| **Turn finely** | Hold `Shift` while dragging. |
| **Back to the default** | Double-click. |
| **Small steps** | Mouse wheel, or click the knob and use the arrow keys (`Shift` for finer). |
| **Give it a hardware control** | MAP mode (chapter 13), or right-click › MIDI Learn. |
| **Use a list (amber text with an arrow)** | Click to open; it opens whole and never runs off the screen. Click a line, or click outside / `Esc` to leave it. |

Selectors are rows of small buttons with an LED over the chosen one. A control printed faintly is not in use at the moment (for example the arpeggiator's Rate while the clock plays).

## FULL SCREEN

`F11`, or **FULL SCREEN** on the FX page, hides the title bar and the Windows taskbar for a distraction-free stage view. **START FULL** (FX page, SETUP) starts Polymorph that way every time. `Esc` is PANIC, so leave full screen with `F11`.

## THE BOTTOM BAR

**PANIC** (red, left), **AUDIO ON** / **AUDIO OFF**, **OUT**, a line naming the MIDI devices in use, a status line, and the name.

# 4 Playing notes

The computer keyboard is a one-octave piano:

| KEY | A | W | S | E | D | F | T | G | Y | H | U | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Note** | C4 | C#4 | D4 | D#4 | E4 | F4 | F#4 | G4 | G#4 | A4 | A#4 | B4 | C5 |

- `Z` and `X` move the keyboard down and up an octave, from −3 to +3. A key you are holding keeps its pitch.
- Computer keys play at velocity 100; velocity comes from pads, keyboards and the sequencer.
- The keys play only while Polymorph is the active window and audio is on.
- **What you play by hand always plays the selected track's sound** (the display says `KEYS → T2`) and goes through the **LIVE channel** of the mixer strip (chapter 7). If the selected track is a DRUM track, the notes play its drums.

## PANIC

The red **PANIC** button, the `Esc` key, or MIDI CC 123 (All Notes Off): every voice and every drum fades out in a few milliseconds, held and latched notes and the arpeggiator are cleared, and the echo and reverb are emptied. The clock keeps running: a playing pattern is heard again from its next step. Press STOP as well to keep it quiet.

## KEYS AT A GLANCE

| KEY | DOES |
|---|---|
| `A` … `K` | Notes C to C |
| `Z` / `X` | Octave down / up |
| `V` **(held)** | FILL (chapter 11) |
| `1`–`4` | Mute track 1–4 |
| `Shift` + `1`–`4` | Solo track 1–4 |
| `PageDown` **or** `.` / `PageUp` **or** `,` | Next / previous pattern |
| `Esc` | Panic |
| `F11` | Full screen on / off |

# 5 The sound of a SYNTH track

Each SYNTH track owns a complete sound. The SYNTH page shows the sound of the track selected with T1–T4.

OSC 1 (FM, RING) + OSC 2 + SUB + NOISE → DRIVE → FILTER LP24 / HP / BP → LOUDNESS CONTOUR

## OSCILLATORS AND MIXER

| CONTROL | RANGE | MEANING |
|---|---|---|
| **OSC 1 waveform** | sine, saw, square, triangle | Changing it while a note sounds crosses over smoothly. |
| **OSC 1, SUB, NOISE** | 0–1 | Levels of oscillator 1, the sub-oscillator (a square one octave below OSC 1) and white noise. |
| **OSC 2 waveform** | the same four | |
| **OSC 2** | 0–1 | Level of oscillator 2 (0 as it comes). |
| **OCTAVE** | −2 to +2 | Of oscillator 2. |
| **SEMITONE** | −12 to +12 | Of oscillator 2. |
| **FINE** | ±50 cents | Of oscillator 2. Small values against OSC 1 give slow beating. |

The four levels share the headroom: while they add up to 1.00 or less they are used as set; above that, all four are scaled down together.

## FM AND RING (COLUMN OSC 2 › 1)

| CONTROL | RANGE | MEANING |
|---|---|---|
| **FM** | 0–100 % | OSC 2 moves OSC 1's phase: frequency modulation. Sidebands grow around every harmonic. The lower half of the knob is the gentle half. |
| **RING** | 0–100 % | Crossfades OSC 1 to OSC 1 × OSC 2. At 100 % the two pitches are gone and you hear their sum and difference. |

- **OSC 2 modulates whatever its own level is.** Turn Osc 2 down to 0 and it is only the modulator.
- **The note stays in tune at any FM amount** (through-zero FM), so FM can be locked per step or thrown by RANDOM without detuning a line.
- The character depends on OSC 2's tuning: same pitch or whole octaves give harmonic tones, from mellow to brassy; a fifth or an odd interval (Semitone 6, 10, Fine off zero) gives bells, metal and clangs. Two sines give the clearest FM.
- With FM far up, bright waveforms are given fewer harmonics so the sound gets hollower rather than gritty. Very high notes with heavy FM can still get a little digital edge.

## THE LADDER FILTER

| CONTROL | RANGE | MEANING |
|---|---|---|
| **CUTOFF** | 20 Hz – 20 kHz | Where the filter works. |
| **EMPHASIS** | 0–1 | Resonance at the cutoff. |
| **CONTOUR** | −1 to +1 | How far the filter contour moves the cutoff. At +1 its peak is 5 octaves above Cutoff; negative values move it down. |

| | | |
|---|---|---|
| **KEY TRACK** | 0–100 % (50 %) | How far the cutoff follows the note, counted from C4. Raise it if high notes sound dull or quiet. |
| **FILTER MODE** | LP 24, HP, BP | In ANALOG · SENDS. LP 24: the 24 dB low-pass. HP: high-pass. BP: band-pass. All three use the same ladder, Emphasis and Contour. |
| **OSC** | on / off | Lets the filter self-oscillate (below). |

### The filter as an oscillator

With **OSC** lit, Emphasis reaches further: from about 87 % of its travel the filter sings by itself, a pure sine at the cutoff.

1. OSC on, Emphasis 1.00, **Key Track 100 %**, Cutoff 262 Hz (middle C), Osc 1 at 0.
2. Play: the keys now play the filter's own tone, in tune.
3. Contour up with a short filter Decay bends its pitch on each note: zaps and kicks.

The tone settles at a fixed, safe level (about a third of an oscillator's, so raise the track Level or use BP, which is louder). Whatever is fed in, the filter's output is rounded off before it can reach a dangerous level.

## CONTOURS

### Filter contour (upper row)

| CONTROL | RANGE | AS IT COMES |
|---|---|---|
| **Attack** | 0.002–2 s | 0.010 |
| **Decay** | 0.002–2 s | 0.300 |
| **Sustain** | 0–1 | 0.30 |
| **Release** | 0.002–3 s | 0.300 |

### Loudness contour (lower row)

| CONTROL | RANGE | AS IT COMES |
|---|---|---|
| **Attack** | 0.002–2 s | 0.010 |
| **Decay** | 0.002–2 s | 0.200 |
| **Sustain** | 0–1 | 0.80 |
| **Release** | 0.002–3 s | 0.300 |

Contour settings are read when a note starts: a change affects the next note, not the one sounding. The filter contour only moves the cutoff when **CONTOUR** is above or below 0.

## ANALOG CHARACTER AND SENDS (ANALOG · SENDS)

| CONTROL | RANGE | MEANING |
|---|---|---|
| **DRIFT** | 0–100 % | Each voice wanders slowly on its own: pitch by up to about 15 cents, cutoff by up to an eighth of an octave. Chords and unisons move against each other. |
| **DRIVE** | 0–100 % | Soft saturation before the filter: more harmonics for the filter to work on, a more even level. |
| **PITCH ENV** | ±24 semitones | Every note starts this far above (or below) its pitch and falls back. |
| **P.DECAY** | 0.005–2 s | How fast it falls back. |
| **ECHO SND, REV SND** | 0–1 | The track's sends to the tape echo and the reverb, for its sequenced notes and for your playing. |

All of these start where they change nothing. Every control of the sound can be locked per step, mapped to a MIDI controller, and is saved with the sound.

# 6 The drum kit of a DRUM track

Any track can be a DRUM track: on the SEQ page, choose **DRUM** for it. It then plays a kit of six synthesized drums and uses none of the eight synth voices.




*The SYNTH page with a DRUM track selected: the DRUM selector and mutes, the chosen drum's knobs, KIT FILTER and KIT MODULATION.*

## THE KIT AND ITS NOTES

| DRUM | NOTE | ALSO ON |
|---|---|---|
| **Kick** | 36 (C2) | 35 |
| **Snare** | 38 (D2) | 40, 37 |
| **Clap** | 39 (D#2) | |
| **Closed hat** | 42 (F#2) | 44, 51 |
| **Open hat** | 46 (A#2) | 49 |
| **Perc / Tom** | 45 (A2) | 41, 43, 47, 48, 50 play it lower or higher, as toms |

These are the standard General MIDI drum notes, so all sixteen pads of a pad controller (notes 36–51) sound. A closed hat cuts off a ringing open hat, as on a real hi-hat. On the computer keyboard press Z three times; then A is the kick, S the snare, E the clap, T the closed hat, U the open hat and H the perc. Harder hits are louder, a little brighter and a little longer.

## MUTING ONE DRUM

Every drum has its own **M** beside it, on the SYNTH page and beside the drum row on the SEQ page (the same mute in both places). A muted drum does not sound, from the sequencer or from a pad; the others play on. A mute stops new hits only: a ringing drum finishes. Drum mutes are saved with the pattern and can be given to a controller button in MAP mode.

## PROGRAMMING DRUMS

1. Switch **REC** on.

2. Choose a drum in the row above the steps: KICK, SNARE, CLAP, CL HAT, OP HAT, PERC.
3. Click steps: each click adds that drum to the step, or takes it off.
4. Choose another drum and click again: **a step can hold several drums**.

You can also hit pads with REC and PLAY (LIVE REC) or with STEP REC: each drum keeps the velocity it was hit with. The step editor's **VELOCITY** knob sets the chosen drum's velocity on that step; Micro, Chance, Cond and Accent act on the whole step.

## SCULPTING A DRUM

With a DRUM track selected, choose a drum in the DRUM module of the SYNTH page; its knobs appear beside it.

| DRUM | ITS KNOBS |
|---|---|
| **Kick** | **TUNE** 30–120 Hz (the pitch it settles on) · **PITCH ENV** (how far above it starts) · **P.DECAY** 5–400 ms (short is a punch, long a "pew") · **CLICK** (the beater's tick) · **DECAY** 0.05–2.5 s · **DRIVE** · **TONE** (how much top is let through) |
| **Snare** | **TUNE** 100–400 Hz · **TONE** (body against noise) · **SNAPPY** (brightness of the noise) · **DECAY** 0.03–1 s |
| **Clap** | **TONE** (where the noise band sits) · **DECAY** 0.05–1.5 s · **SPREAD** (the gap between the hand bursts) |
| **Closed hat** | **TUNE** ×0.5–×2 (six metallic oscillators) · **TONE** · **DECAY** 10–400 ms |
| **Open hat** | **TUNE** · **TONE** · **DECAY** 0.05–2 s |
| **Perc / Tom** | **TUNE** 60–800 Hz · **PITCH ENV** · **DECAY** 0.03–1.5 s |
| **Every drum** | **LEVEL** 0–1.25 · **PAN** · **ECHO SEND** · **REV SEND** |

### Starting points for the kick

| FOR | TUNE | PITCH ENV | P.DECAY | CLICK | DECAY | DRIVE | TONE |
|---|---|---|---|---|---|---|---|
| **Punchy techno kick** | 48 Hz | 70 % | 40 ms | 40 % | 0.30 s | 40 % | 70 % |
| **Hard, driven kick** | 55 Hz | 85 % | 25 ms | 70 % | 0.22 s | 80 % | 85 % |
| **Long, deep boom** | 42 Hz | 35 % | 60 ms | 5 % | 1.8 s | 0 % | 30 % |

## KIT FILTER AND KIT MODULATION

| CONTROL | MEANING |
|---|---|
| **Kit Filter: CUTOFF, EMPHASIS** | The ladder filter over the whole kit (low-pass). Fully open with no Emphasis it is out of the path. |
| **Kit Filter: DRIVE** | Saturation of the whole kit before that filter. |
| **Kit Modulation, two routes** | Source (lfo1, mod wheel, velocity) › any drum setting › Amount (−1 to +1). Read at the moment a drum is hit. |

Examples: `velocity › chh_decay` +0.3 makes hard hat hits longer; `mod wheel › kick_decay` lets the wheel open the kick; `lfo1 › snare_tone` moves the snare slowly (raise the LFO's Depth).

**Locks work for drums too:** with REC on, right-click a step to hold it and turn a drum knob (chapter 10). A longer kick on step 13, a snare tuned up on one hit, a hat that opens on one step. **RANDOM** works for a kit as well (chapter 8).

A DRUM track shares with a SYNTH track: Level, M, S, meter, length, speed, conditions, chance, microtiming, locks, patterns and files. Of the timed events, only NOTE (an extra drum hit inside the step) acts on a DRUM track.

# 7 The mixer strip and the LIVE channel

| PART | WHAT IT DOES |
|---|---|
| **T1–T4** | Select the track: the SYNTH page shows its sound or kit, the SEQ page its steps, and keys and pads play it. The LED is lit while the track is audible. |
| **M / S** | Mute / solo the track (keys `1`–`4`, `Shift`+`1`–`4`). |
| **LEVEL** | The track's volume, 0–1.25 (1.00 as it comes). It also scales what the track sends to the effects. |
| **DUCK** | How much the sidechain turns this track down at each trigger (chapter 12). |
| **LIVE** | The channel of **what you play by hand** (keys, pads, arpeggiator). Under it, `→ T2`: the track whose sound you are playing, always the selected one. |
| **LIVE M** | Mutes your playing; the sequencer plays on. |
| **LIVE LEVEL** | The volume of your playing, 0–1.25. It never changes a track's level. |
| **LIVE ECHO, REVERB** | Extra echo and reverb for your playing only, added to the sound's own sends. |
| **MOD** | An on-screen mod wheel; it follows a hardware wheel or X-Y pad. |
| **MASTER** | The output volume. 0.50 puts one saw note at about −10 dBFS, 1.00 at about −3 dBFS. |
| **Master meters** | Left and right after the limiter; the top two segments are red, close to full scale. |
| **FX ON / MONO** | Echo and reverb in or out; a mono check of the whole mix (chapter 15). |

## MIXING YOUR PLAYING AGAINST THE PATTERN

1. Let the pattern play.
2. Select the track whose sound you want under your hands (T1–T4).
3. Balance with **LIVE Level** against the track Levels. Your notes use the track's sound but not its Level.
4. Your notes get the sound's own Echo Snd and Rev Snd, like its sequenced notes; LIVE Echo and Reverb add more, for your playing only.

## MUTE AND SOLO

- Muting acts at once: no new notes; a sounding note ends normally.
- Unmuting waits for the track's step 1, so the track comes back at the start of its pattern.
- While any track is soloed, only soloed tracks play.
- Mutes and solos belong to the instrument, not to a pattern.

## VOICES

**VOICES** (FX page, SETUP): **1** is monophonic; **4 or 8** give each held note its own voice. When all are busy, a new note first takes the quietest voice whose key was released, otherwise the note that has sounded longest. The keys, MIDI, the arpeggiator and the SYNTH tracks share the voices; DRUM tracks use none. A note is equally loud with 1, 4 or 8 voices; chords add up.

# 8 LFO, modulation and RANDOM

## THE LFO

There is one LFO for the whole instrument (MODULATION · LFO on the SYNTH page).

| CONTROL | RANGE | AS IT COMES |
|---|---|---|
| **WAVE** | SINE, TRIANGLE, SQUARE, S&H, RANDOM | SINE |
| **SYNC** | FREE, 1/1, 1/2, 1/4, 1/8, 1/16, 1/32 | FREE |
| **RATE** | 0.05–20 Hz | 5 Hz |
| **DEPTH** | 0–1 | 0 |

- **S&H** (sample and hold): a new random level each cycle, held until the next. On pitch: bleeps. On cutoff: a different brightness each step.
- **RANDOM**: the same random levels joined by a smooth curve: wandering.
- **Sync**: with a division, the LFO makes one cycle per division at the tempo and stays locked to the song: S&H with Sync 1/16 gives a new level on every sixteenth. Rate is then printed faintly.
- A pattern played from its start gets the same line of S&H levels every time.

## THE MOD MATRIX

Two slots, each a **source**, a **destination** and an **AMOUNT** from −1 to +1.

| SOURCES | DESTINATIONS |
|---|---|
| **lfo1, mod wheel, velocity, none** | pitch (1.00 = 12 semitones) · cutoff (1.00 = 5 octaves) · amplitude (only lowers) · lfo depth (adds to Depth) · echo rate (the tape echo's speed, 1.00 = an octave either way) |

The LFO only reaches a destination as far as its depth allows: with Depth 0 and nothing feeding `lfo depth`, LFO routes are silent. As it comes, slot 1 is `mod wheel › lfo depth` 1.00 and slot 2 `lfo1 › pitch` 0.02: the mod wheel brings in a gentle vibrato.

## RANDOM: SOMETHING DIFFERENT ON EVERY STEP

Each track has its own RANDOM (SOUND · RANDOM, bottom right of the SYNTH page): a list of **what** is thrown, and an **AMOUNT**. On every step the sequence plays, that one setting is thrown off by a new random amount, for that step only. The knob itself does not move, and notes played by hand are not thrown.

| SYNTH TRACK | EACH STEP GETS |
|---|---|
| **PITCH** | Moved by whole semitones, up to 12 either way |
| **CUTOFF** | Brighter or darker |
| **DECAY** | Shorter or longer |
| **PAN** | A place between left and right |
| **ECHO / REV SEND** | More or less send |
| **FM, VELOCITY** | More or less |

| DRUM TRACK | EACH STEP GETS |
|---|---|
| **TUNE, TONE, DECAY, PAN, sends** | That setting of each drum the step hits |
| **VELOCITY** | Harder or softer hits |

- **Amount**: at 100 % a control can land anywhere within half its range either side of its knob; 20–50 % is the lively middle.

- **The dice belong to the pattern.** Each pattern keeps a seed, so it throws the same values on the same steps every time it plays from the start, also after saving and loading. Random to the ear, repeatable in fact.
- **DICE** throws new dice (another seed) for the pattern on view; the display shows the seed.

# 9 Clock and arpeggiator

## THE MASTER CLOCK

| CONTROL | MEANING |
|---|---|
| **TEMPO** | 30–300 BPM. |
| **PLAY** | Start, or continue from where it stopped. |
| **STOP** | Pause the clock; sequenced notes are released. |
| **RESET** | Back to bar 1, step 1. **Nothing is erased**; it keeps playing if it was playing. |
| **INT / EXT** | Its own tempo, or follow MIDI clock (Clock, Start, Stop, Song Position) from any input. With EXT, play, stop and tempo come from the other device. |

The clock drives the sequencer, the arpeggiator while the clock plays, the synced LFO, and the tape echo when its time comes from a division. It can also be sent out to other gear (chapter 13).

## THE ARPEGGIATOR

Hold a chord: the arpeggiator plays its notes one at a time, through the LIVE channel, with the selected track's sound.

| CONTROL | MEANING |
|---|---|
| **ARP** | On / off. |
| **LATCH** | Keeps going after you let go; the next chord replaces it. |
| **MODE** | up, down, up down, as played, random. |
| **RATE** | 0.5–40 steps per second. In use while the clock is **stopped**. |
| **DIVISION** | 1/4, 1/8, 1/16, 1/32 and triplets. In use while the clock is **playing**. Can be chosen before pressing PLAY. |
| **GATE** | 0.05–1.00: how much of each step the note lasts. |
| **OCTAVES** | 1–4: repeats the chord in higher octaves. |

The one of Rate and Division not in use is printed faintly, and a small amber line says `FREE 8.0/s` or `CLOCK 1/16`. The arpeggiator is not swung.

# 10 The sequencer

Four tracks, each up to 64 steps. A step is a sixteenth note at Speed 1x and holds its own notes, velocity, length, timing, chance, condition, accent, slide, parameter locks and timed events.

## REC DECIDES: LOOKING, OR CHANGING

**A step can only be changed while REC or STEP REC is lit.** With both off, the pattern is safe: play, listen and shape the sound, and nothing is written. Notes are never typed: they come from the keys or a controller.

| | REC AND STEP REC OFF | REC ON, CLOCK STOPPED: GRID REC | REC ON, CLOCK RUNNING: LIVE REC | STEP REC |
|---|---|---|---|---|
| **Playing keys or pads** | Plays the selected track's sound | Plays, and is remembered for the next click | Is recorded in time | Each note or chord goes into the selected step, which then moves on |
| **Click a step** | Shows its settings (and plays it once if stopped) | Trig on or off, with the last note or chord played | The same | Chooses where the next note goes |
| **Right-click a step** | Selects it silently | **Holds** it: knobs now set this step only | The same | The same |
| **Step editor, A/S marks, CLEAR** | Do nothing (REC IS OFF) | Work | Work | Work |

While recording, REC is lit red, a red frame goes round the steps, and a red line over them says which mode is on. Only one kind of recording is on at a time.

**THE WORKFLOW**

1. **REC off.** Play a note, listen, turn knobs until the sound is right. Click steps to hear what they hold.
2. **REC on** (clock stopped). Click the steps that should have the note you played last. Click a lit step to remove it.
3. **REC off**, find the next note or sound; **REC on**, assign it. Repeat.
4. For a different sound on one step only: REC on, **right-click the step**, turn knobs, right-click again to let go.

## THE STEP EDITOR: EIGHT KNOBS

| KNOB | RANGE | MEANING |
|---|---|---|
| **NOTE** | C–1 to G9 | The step's lowest note; turning transposes the whole step. On an empty step it puts a note there. |
| **VELOCITY** | 1–127 | How hard the step's notes are played. |
| **LENGTH** | 0.05–4.00 steps | Above 1.00 the note rings across the next steps. |
| **MICRO** | −12 to +12 ticks | Early or late, up to half a step (chapter 11). |
| **CHANCE** | 0–100 % | Chance of playing on each pass (chapter 11). |
| **COND** | stepped | When the step is allowed to play (chapter 11). |
| **ACCENT** | OFF / ON | Louder and brighter. |
| **SLIDE** | OFF / ON | Glides into the next step (SYNTH tracks). |

An amber line over the knobs says the step's rule in words, and the display says it while you turn Chance or Cond.

## ACCENT AND SLIDE

Under each step button are two small marks, **A** and **S**: click to switch them, or use the step editor's knobs.

| | WHAT IT DOES | HOW MUCH |
|---|---|---|
| **Accent** | The notes are louder and their filter opens further (more Contour). A drum is hit harder. | **ACCENT** knob, FX page, GROOVE: 0–100 % (60 %) |
| **Slide** | The note is held into the next step, whose note has **no new attack**: the pitch glides to it and the envelopes carry on (legato). SYNTH tracks only. | **SLIDE** knob, GROOVE: glide time 10–500 ms (60 ms) |

A slide needs a next step with a note; several slides in a row make one long legato phrase. For an acid line: saw, Cutoff low, Emphasis high, Contour up, short filter Decay; accents on a few steps, slides before some of the jumps.

## PARAMETER LOCKS

1. Switch **REC** on.
2. **Right-click the step.** The display says `HOLD STEP 05`.
3. Turn any knob of the sound or the kit (on screen or on a mapped controller), or pick a waveform: the value is written for that step only. The step shows an **L**.
4. Right-click again to let go (or switch REC off).

**With a hardware controller:** press and keep the mouse button down on the step while you turn the hardware knob, then let go. While a step is held, the SYNTH page shows it as it will sound; the STEP LOCKS list names each lock, and **CLEAR LOCKS** removes them. Everything of a sound can be locked except LFO rate and shape. Effects, tempo, the mixer and the drum mutes are not part of a step.

## TIMED EVENTS: THINGS THAT HAPPEN INSIDE A STEP

Up to 16 per step, each a set time after the step starts.

| KIND | WHAT IT DOES | VALUE |
|---|---|---|
| **NOTE** | An extra note (the note you played last) | velocity 1–127 |
| **OFF** | Ends a note early | — |
| **PARAM** | Changes a control of the sound for this step (chosen in the list beside ADD) | the control's whole range |
| **TRANSP** | Shifts the sounding notes | −48 to +48 semitones |

Choose the kind, set **OFFSET** (0–1000 ms) and **VALUE**, press **ADD**. Pick an event in the list to edit it; **DEL** removes it. Events move with the step's microtiming and happen only when the step plays.

## TRACKS

| CONTROL | MEANING |
|---|---|
| **T1–T4** | Which track the steps, the step editor and the SYNTH page show. |
| **SYNTH / DRUM** | What the track is. Switching keeps the steps; the track remembers both its sound and its kit. |
| **LENGTH** | 1–64 steps. |
| **SPEED** | 1/8, 1/4, 1/2, 1x, 2x of the tempo. |
| **SWING** | The track's own swing (chapter 11). Fully down (`GLB`) it follows the global Swing. |
| **1-16 … 49-64** | Which 16 steps the step buttons show. |

Each of the 16 patterns has its own four sounds and kits; copying a pattern copies them. Shared by the whole instrument: Master, Voices, the clock, the effect settings (each track has its own sends), the LIVE channel, and the LFO's rate and shape.

## SIXTEEN PATTERNS

- Click a button 1–16 (or `PageDown` / `PageUp`, `.` / `,`) to play that pattern. While the clock runs, **the change happens exactly on the next bar line**; when stopped, at once.
- **EDIT ONLY**: a click shows another pattern for editing while the current one keeps playing.
- **COPY TO** N copies the pattern on view into slot N.
- Under the buttons: which pattern plays, which you edit, and the pattern's name, with *(not saved)* while it has changes.

## RECORDING BY PLAYING

**LIVE REC**: select the track, press **REC** and **PLAY** (either order), and play. Each note goes to the nearest step with its own velocity, timing and length.

- **QUANTIZE** on: notes sit exactly on the step. Off: your timing is kept (as microtiming).
- **OVERDUB** on: notes are added to what the step holds. Off: the first note on a step replaces it.
- Several notes on one step each keep their own velocity and timing; what plays back is what you played. Knob movements are not recorded.

**STEP REC**: with the clock stopped, press **STEP REC**, click the step to start at, and play: each note or chord is written there and the selection moves to the next step. On a DRUM track each pad is written as its drum.

## RESET AND THE CLEAR BUTTONS

| BUTTON | WHAT IT DOES | ERASES |
|---|---|---|
| **RESET** | Clock back to bar 1, step 1; restarts the chance and condition counting | Nothing |
| **STOP** | Pauses the clock | Nothing |
| **PANIC** | Silences everything at once | Nothing |
| **CLEAR STEP** | Empties the selected step: notes, locks, events, accent, slide | That step |
| **CLEAR LOCKS** | Removes the selected step's locks; notes stay | Its locks |
| **CLEAR** | Empties the whole pattern on view (no question asked) | That pattern |

# 11 Sequencer guide: conditional trigs

A step normally plays every time its track comes round. Chance and conditions decide *whether* it plays this time; microtiming and swing move it in time. Together they turn one bar of steps into music that changes over many bars.

> **TRY IT WITH THE LESSON BANK**
> Press **LOAD BANK** and open `demos\learn_conditions.json`, press PLAY, and click through patterns 1–7. In each one, track 1 is a plain kick and hat to count against; the lesson is on tracks 2 and 3. With REC off, click the steps named below to see their settings.

## THE RULES AT A GLANCE

| SETTING | MEANS | THE DISPLAY SAYS |
|---|---|---|
| **Chance 50 %** | Toss a coin every time the step comes round | `50% · plays about 50 times in 100` |
| **Cond 1:2** | Count the track's loops in twos; play on the first | `1:2 · plays on loop 1 of every 2` |
| **Cond 2:2** | Count in twos; play on the second | `2:2 · plays on loop 2 of every 2` |
| **Cond 3:4** | Count in fours; play on the third | `3:4 · plays on loop 3 of every 4` |
| **PRE** | Play only if the last step with a rule did play | `PRE · plays if the step before it did` |
| **NOT PRE** | Play only if that step did not play | `NOT PRE · plays if the step before it did not` |
| **FILL** | Play only while FILL is held (button or V) | `FILL · plays only while FILL is held` |
| **Cond + Chance** | First the condition, then the coin | `1:2 · … then 50% of the time` |

A **loop** is one pass of **that track**: a 12-step track counts its own loops. **RESET** starts the count again at loop 1.

## HOW A:B COUNTS

Read **A:B** as "play on loop A of every B". B can be 2 to 8.

| CONDITION | LOOP 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **none** | ● | ● | ● | ● | ● | ● | ● | ● |
| **1:2** | ● | ○ | ● | ○ | ● | ○ | ● | ○ |
| **2:2** | ○ | ● | ○ | ● | ○ | ● | ○ | ● |
| **1:4** | ● | ○ | ○ | ○ | ● | ○ | ○ | ○ |
| **2:4** | ○ | ● | ○ | ○ | ○ | ● | ○ | ○ |
| **3:4** | ○ | ○ | ● | ○ | ○ | ○ | ● | ○ |
| **4:4** | ○ | ○ | ○ | ● | ○ | ○ | ○ | ● |
| **1:8** | ● | ○ | ○ | ○ | ○ | ○ | ○ | ○ |
| **Chance 50 %** | ◐ | ◐ | ◐ | ◐ | ◐ | ◐ | ◐ | ◐ |

● plays · ○ silent · ◐ a coin toss each time

## LESSON 1: CHANCE 50 %




Track 2: filled = always (C3) · outlined = Chance 50 % (E3)

The C never fails; the E between the beats comes and goes with no pattern. Raise Chance on step 3 and it comes more often.

## LESSON 2: EVERY OTHER LOOP (1:2 AND 2:2)

Track 2 plays a rising phrase on steps 1, 4, 7, 11 with **1:2**; track 3 a falling answer on the same steps with **2:2**. Listen for: question, answer, question, answer: a two-bar phrase out of one bar of steps.

## LESSON 3: ONE LOOP IN FOUR (1:4 … 4:4)

| TRACK | STEP | NOTE | COND | PLAYS ON LOOP |
|---|---|---|---|---|
| **2** | 1 | C3 | 1:4 | 1, 5, 9 … |
| **2** | 5 | E3 | 2:4 | 2, 6, 10 … |
| **3** | 9 | G4 | 3:4 | 3, 7, 11 … |
| **3** | 13 | C5 | 4:4 | 4, 8, 12 … |

One note per bar, a different one each bar, climbing over four bars and starting again.

## LESSON 4: PRE AND NOT PRE

| STEP | NOTE | SETTING | WHAT HAPPENS |
|---|---|---|---|
| **1 (and 9)** | C3 | Chance 50 % | The coin toss |
| **3 (and 11)** | C4 | PRE | Plays only when step 1 (9) played: an answer above |
| **4 (and 12)** | G2 | NOT PRE | Plays only when it did not: a low note instead |

You hear either “low C, high C”, or a single G below. PRE and NOT PRE look back at the last step that has a Chance or condition of its own; they do not count as such a step themselves, so several in a row all look at the same one.

## LESSON 5: FILL

A steady bass note on steps 1, 5, 9, 13; a rising run on 3, 7, 11, 14, 15, 16 with **FILL**, and a snare roll on drum steps 13–16 with **FILL**. Hold V (or the FILL button) during the last bar of a phrase and let go on the one.

## LESSON 6: MICROTIMING

A step has 24 ticks; **MICRO** goes from −12 (half a step early) to +12 (half a step late). In the lesson, steps 3, 7, 11, 15 sit at +8, a third of a step late, against a straight hi-hat: a swing. Set them back to 0 and the line straightens.

## LESSON 7: POLYMETER 16 · 12 · 7

| TRACK | LENGTH | PLAYS |
|---|---|---|
| **1 (drums)** | 16 | The bar |
| **2** | 12 | A four-note bass figure that starts again every three beats |
| **3** | 7 | Three bell notes that start again every seven steps |

The lines slide against each other: the bass meets the bar line every three bars, the bell every seven, all three together every 21 bars. Change a track's **LENGTH** to hear the cycle change. **SPEED** does the same another way: a track at 1/2x takes two

bars over its 16 steps.

## SWING

Swing delays every second step of the grid, so pairs of notes go long–short.

| CONTROL | WHERE | MEANING |
|---|---|---|
| **SWING** | Top strip | Global, 50–75 %. 50 % straight, 58 % a light shuffle, 66 % a triplet feel, 75 % the dotted extreme. |
| **SWING (track)** | SEQ page, TRACKS | The track's own. Fully down (GLB) it follows the global knob; at 50 % it is always straight. |
| **SWING GRID** | FX page, GROOVE | 1/16: every second sixteenth is late (usual). 1/8: every second eighth. |

Swing is exact to the sample and comes on top of Micro. It follows the song's grid, so tracks at other speeds swing together. Keep the kick track's own Swing at 50 % for a dead-straight kick under a shuffled groove.

> **IDEAS TO MAKE ONE BAR LAST A WHOLE SET**
>
> - Open hat with 1:2 on the off-beats, a ghost snare with Chance 30 %.
> - A crash or long note with 1:8 on step 1: once every eight bars.
> - A bass note with Chance 50 % followed by PRE / NOT PRE answers: call and response that never repeats the same way.
> - Hats on a 15-step track against a 16-step kick: the accent drifts across the bar.

# 12 Effects




*The FX page: tape echo, reverb, rumble and sidechain duck; below them SETUP, GROOVE and MIDI.*

## FX ON AND SENDS

**FX ON**, beside Master in the mixer strip, switches echo and reverb in (with a 20 ms fade; switching back on starts with an empty tape). All sends start at 0, so the sound stays dry until you raise one.

| WHERE | WHAT IT FEEDS |
|---|---|
| **Echo Snd, Rev Snd (ANALOG · SENDS)** | The track's notes, sequenced and played by hand |
| **Echo Send, Rev Send of each drum** | That drum |
| **LIVE Echo, Reverb (mixer strip)** | Extra, for your playing only |

## TAPE ECHO

Modelled on a loop of tape passing a record head and three playback heads.

| CONTROL | RANGE | MEANING |
|---|---|---|
| **REPEAT RATE** | 0.050–0.700 s | Time to head 1. It is the tape speed: moving it bends the pitch of what is on the tape until the motor settles. |
| **TIME FROM** | Repeat Rate / clock division | With a division the time follows the tempo (1/4 … 1/32T). |
| **HEADS** | 1, 2, 3, 1+2, 2+3, 1+3, ALL | Head 2 repeats at twice the time, head 3 at three times. Head 1 leans left, head 3 right. |
| **INTENSITY** | 0–1 | How much echo is recorded again. Above about 0.85 it sustains itself into a loud, saturated tone that stays bounded. |
| **WOW** | 0–1 | Slow drift and fast shimmer of the tape speed. |
| **LEVEL** | 0–1 | Loudness of the echoes. |
| **TO REV** | 0–1 | How much of the echoes also goes into the reverb. |

| | | |
|---|---|---|
| **HOLD** | on / off | Freezes the tape as a loop (below). |

### HOLD and pitch dives

- With **HOLD** lit, nothing new is recorded and the tape goes round as it is: the repeats neither die nor build up. The loop is as long as the furthest head heard. Going in and out is click-free.
- **Pitch dives:** set Time from to Repeat Rate, map Repeat Rate to a fader (MAP mode), hold a phrase, and move the fader: towards 700 ms the sound sinks, towards 50 ms it shoots up. The motor has a little inertia, so the pitch glides.
- **LFO on the tape:** in the mod matrix, `lfo1 › echo rate`. Small amounts (0.05–0.25) give a seasick tape; an S&H LFO makes it jump.

## REVERB

| CONTROL | RANGE | MEANING |
|---|---|---|
| **PRE-DLY** | 0–0.200 s | Gap before the reverb starts. |
| **DECAY** | 0.2–10 s | Time for the tail to fall 60 dB. |
| **DAMPING** | 0–1 | How fast the treble of the tail dies. |
| **LEVEL** | 0–1 | Loudness of the reverb. |

## SIDECHAIN DUCK

At each trigger the ducked tracks dip and come back: the pumping of a pad or bass under the kick.

| CONTROL | MEANING |
|---|---|
| **DUCKED BY** | The trigger track (T1–T4) or OFF. A DRUM track triggers with its kick; a SYNTH track with every note it starts. |
| **DEPTH** | How far down, 0–100 % (80 %). |
| **ATTACK** | How fast it goes down, 1–50 ms (5 ms). |
| **RELEASE** | How long it takes to come back, 30 ms – 1 s (250 ms). |
| **DUCK (mixer)** | How much each track takes. At 0 (as it comes) a track is not ducked. |

Usual set-up: kick on T1, Ducked by T1, DUCK up on the bass and pad, at 0 on the drums. The dip is always a smooth curve. Sends dip with the track; the LIVE channel is not ducked.

## RUMBLE

The deep tail under a techno kick: one drum feeds a dark, low reverb of its own, which ducks under each hit of that drum so the kick stays clear.

| CONTROL | MEANING |
|---|---|
| **FROM THE DRUM** | Which drum feeds it (the kick as it comes). |
| **AMOUNT** | How much goes in. **At 0 the rumble is off.** |
| **DECAY** | The tail's length, 0.2–4 s. |
| **TONE** | Its low-pass, 40–400 Hz: lower is darker and deeper. |
| **LEVEL** | Its loudness. |

Nothing below 25 Hz passes and the rumble's output is held below full scale, so it cannot build up without limit.

# 13 MIDI

Everything about MIDI is in the MIDI module on the FX page: three inputs, one output, and the mapping tools.

## THREE INPUTS AND THEIR ROLES

**IN 1**, **IN 2** and **IN 3** each have a device list and three role buttons:

| ROLE | FOR | WHAT IT DOES |
|---|---|---|
| **NOTES** | A keyboard or pads that only play | Notes play; pitch bend (±2 semitones), mod wheel (CC 1) and sustain (CC 64) work. |
| **CTRL** | A box of faders, knobs and buttons | Controls move knobs and buttons do actions. Its notes never play. |
| **BOTH** | A keyboard with knobs; pads with an X-Y pad | It plays, and its mapped controls act too. |

1. Connect the devices (press **REFRESH** if they were connected after Polymorph started).
2. Choose a device for each input you need. A device can be on one input only.
3. Press AUDIO ON: ports open when audio starts.

Every device keeps its own map and role under its name.

## READY-MADE MAPS

When a device is chosen for the first time, Polymorph looks at its name and gives it a starting map. The map found is shown in MAP mode.

| DEVICE NAME CONTAINS | ROLE | STARTING MAP |
|---|---|---|
| **nanoPAD** | BOTH | Pads: notes on channel 1 with velocity (drums on a DRUM track). X-Y pad left–right (CC 1): mod wheel; up–down (CC 2): Cutoff of the selected track. |
| **nanoKONTROL** | CTRL | Faders 1–4: track levels; fader 5: LIVE; fader 8: Master. Knobs 1–4: Cutoff, Emphasis, Contour, Key Track. S and M buttons 1–4: solo and mute. Play, stop, record. |
| **Arturia, KeyStep, MiniLab, KeyLab** | BOTH | Keys, wheel, bend; common knob numbers to the filter, contour, sends and Master. |
| **SP-404** | BOTH | Its pads play; link its controls in MAP mode. |
| **anything else** | BOTH | It plays; CC 74 / 71 / 73 move Cutoff / Emphasis / Attack. |

## ANY FADER TO ANY KNOB

A hardware control is linked by **what it sends**, not by its shape, so any fader or knob can drive any knob: a fader to a track Level, to Master, to a send, to Cutoff.

- Polymorph finds out by itself whether a control sends **positions** (a knob with two ends, a fader) or **steps** (an endless encoder). MAP mode shows `POSITIONS` or `STEPS`; press `R` if it guessed wrong.
- Positions travel the knob's whole range on the knob's own curve: hardware at half way is the knob at half way. Steps move the value from where it is and never jump.
- The on-screen knob moves with the hardware; moves are smoothed (about 20 ms), so sweeps have no zipper noise. Faders that send pitch bend work too.

## TAKEOVER

| TAKEOVER | WHEN THE HARDWARE AND THE SCREEN DISAGREE |
|---|---|
| **DIRECT** | The value goes to where the hardware is at once. It follows your hand; the first touch can jump. |
| **PICK-UP** | Nothing happens until the hardware reaches the value; then it takes over. No jumps. The display says `PICK-UP: TURN UP` or `TURN DOWN` meanwhile. |

## MAP MODE

Press **MAP** in the top strip. Every linkable control gets an amber frame and a tag with its controller ( `CC16` , `FDR3` , `N51` ). While MAP is on, a click picks a control instead of using it.

| TO | DO |
|---|---|
| **Link a control** | Click it on screen, then move the hardware knob or fader, or press the button or pad. |
| **Link another** | Click the next one. A hardware control does one job: giving it a new one takes it from the old. |
| **Remove a link** | Click the control, press `Delete` . |
| **Set a range** | The display shows `MIN 0% · MAX 100%` . Mouse wheel over the left half of the display for MIN, right half for MAX (or the arrow keys), in 5 % steps. MIN above MAX turns the control round. |
| **Fix it to a track** | The line ends in `TRACK SEL` (follows the selected track). Click it or press `T` for TRACK 1 … TRACK 4. |
| **Leave** | Press MAP again. |

Everything can be linked: every knob, and every button that acts by itself (PLAY, STOP, REC, PANIC, M and S, patterns, ARP, LATCH, FX ON, MONO, HOLD, LIVE M, each drum's M). A pad can become a button: click PLAY, hit the pad; that pad then starts the clock instead of playing a note. A button acts when its controller sends 64 or more. **Linking never interrupts the sound.**

**SAVE MAP** / **LOAD MAP** keep named mapping sets in `user\mappings` , one per performance. With **WITH BANK** lit, banks save and load their links too. Outside MAP mode, right-click a control › **MIDI Learn** links one control the same way; **Forget MIDI control** removes it.

## MIDI CLOCK OUT

**OUT** in the MIDI module chooses a MIDI output; **SEND CLOCK** switches the clock on. While audio is on, Polymorph sends clock (24 per quarter note, also while stopped), Start on PLAY from the beginning and on RESET while playing, Stop on STOP, and Song Position plus Continue when playing on from where it stopped. Each clock belongs to an exact sample of the sequencer, so other gear follows tightly (chapter 15).

## LIVE ACTIONS ON THE CONTROL CHANNEL

MIDI channel **16** is for buttons, not for playing: notes on it never sound.

| ON CHANNEL 16 | ACTION |
|---|---|
| **Notes 36–39 (C2–D#2)** | Mute track 1–4 (again to unmute) |
| **Notes 40–43 (E2–G2)** | Solo track 1–4 |
| **Note 44 / 45** | Previous / next pattern |
| **Note 47** | Panic |
| **Notes 48–63** | Pattern 1–16 |
| **Program Change 1–16** | Pattern 1–16 |

# 14 Saving and loading

SAVE / LOAD **PATTERN**, **BANK** and **PROJECT** are in PATTERNS · FILES on the SEQ page; SAVE / LOAD **SOUND** in SOUND · RANDOM on the SYNTH page; SAVE / LOAD **MAP** in MIDI on the FX page.

| BUTTON | SAVES / LOADS | FOLDER |
|---|---|---|
| **SAVE PATTERN** | The pattern on view: four tracks with steps, sounds, kits and levels, its dice, the tempo | `user\patterns` |
| **LOAD PATTERN** | A pattern **into the selected slot**; the other fifteen are untouched | |
| **SAVE SOUND** | The selected track's sound or drum kit (no notes) | `user\sounds` |
| **LOAD SOUND** | A sound or kit into the selected track; its notes stay | |
| **SAVE BANK** | All 16 patterns, the tempo and the groove tools (swing, accent, slide, duck, rumble) | `user\banks` |
| **LOAD BANK** | All 16 patterns | |
| **SAVE / LOAD MAP** | A set of controller links | `user\mappings` |
| **SAVE PROJECT** | **The whole set**: 16 patterns, controller links, effect settings, groove tools, Master, LIVE, mutes and solos, arpeggiator, tempo, which pattern plays and which track is selected | `user\projects` |
| **LOAD PROJECT** | All of that back, as it was | |

> **BEFORE A GIG**
>
> **Save a project.** Load it and the instrument is exactly as you left it. What a project does not hold belongs to the computer (output, sound system, block, MIDI devices), and is remembered by itself.

- The display confirms: `SAVED: my groove`, `LOADED: my groove`.
- Unsaved changes show as *(not saved)* and a `*` on the display. Polymorph asks before closing or loading over unsaved work.
- While the clock plays, a loaded pattern, bank or project takes over at the next bar line; nothing stops.
- The wrong kind of file is not loaded; the display names the right button (`THAT IS A BANK FILE`).
- Files are readable JSON in a permanent format: later versions will read them. Older files are converted as they are read (`· MIGRATED`); nothing changes on disk until you save. A damaged file never stops the program: what can be read is loaded.

## AUTOSAVE AND CRASH RECOVERY

Every two minutes, and when you close the program, the whole set is saved as a project into `user\autosave` (the newest five are kept). Saving never makes the audio wait. If Polymorph was ended without closing (a crash, a power cut), it asks at the next start whether to restore the last autosave.

# 15 Playing live

## HEALTH

Top right of the display, while audio is on: `PEAK 43% · LOST 0 MS · DROPS 0`.

| PART | MEANING | GOOD |
|---|---|---|
| **PEAK** | The slowest audio block of the last 10 seconds: how much of its time the computer needed. 100 % means only just in time. | Under about 70 % |
| **LOST** | Milliseconds of audio that did not reach the sound card in time since AUDIO ON. Polymorph measures this itself. | **0** |
| **DROPS** | How many times that happened. | **0** |

If LOST or DROPS move during a set: Block 1024, Card buffer HIGH, or LIGHT LAPTOP. Every session also writes a log, one line per second, to `diagnostics\sessions`; if something went wrong in a set, it says when.

## THE SOAK TEST

Before trusting a computer on stage, let Polymorph play a busy scene on it for an hour and judge the run. Open PowerShell in the Polymorph folder and type:

```
.\Polymorph.exe --self-test       a one-minute check that everything starts
.\Polymorph.exe --soak 60         60 minutes with your audio settings
.\Polymorph.exe --soak 60 light   60 minutes with the LIGHT LAPTOP settings
.\Polymorph.exe --soak perform    no scene: you play, it logs
```

Add `silent` to run without hearing it, or `full` for full screen. The scene uses four tracks, drums, FM, RANDOM, the singing filter, echo, reverb, duck, rumble, pattern changes and clock out. The report (`report.md`, with **PASS** or **FAIL** and three graphs) is written to `diagnostics\sessions\soak-<date-time>`. It passes when **no audio is lost**, **no block reaches 90 %** of its time and **memory grows less than 5 %**. Run it on the computer, output and power supply you will play with.

## LIGHT LAPTOP

One button in SETUP for a weaker computer: 4 voices, Block 1024, Card buffer HIGH, the display redrawn half as often. It roughly halves the processor load. The price: at most four notes at once, and pads answer later (about 100 ms instead of 30). Press it again to go back.

## PA SAFETY

- **Nothing under 20 Hz leaves the instrument:** a high-pass on the master output removes DC and subsonic energy that would only heat a PA's amplifiers (less than 0.3 dB change at 40 Hz).
- **MONO** (under FX ON) sends left + right to both sides, as a mono PA would. Use it to check that echo heads, drum pans and random pan still work, then switch it off.
- The output limiter (ceiling just under full scale) is always on. If the master meters stay in the red, lower Master or the track levels.
- Keep Polymorph's peaks around −6 dB and turn up on the mixer, not in the instrument.

## SYNCING OTHER GEAR

Polymorph can be the **master clock** of a small setup:

**POLYMORPH · CLOCK OUT** → USB → **SAMPLER (FOLLOWS USB CLOCK)** → MIDI out → **SYNTHESIZER (FOLLOWS MIDI CLOCK)**

1. In Polymorph (FX page, MIDI): choose the sampler as **OUT** and light **SEND CLOCK**.
2. On the sampler: set its MIDI sync to follow the USB clock, and switch on its MIDI sync out so it passes the clock on.
3. Connect the sampler's MIDI out to the synthesizer's MIDI in (a TRS-to-DIN MIDI cable if the sampler has 3.5 mm jacks), and set the synthesizer to follow external clock.
4. Press PLAY in Polymorph: everything starts together and follows its tempo.

The other machines get tempo and start/stop; each plays its own patterns. Route their audio into the sampler or a mixer alongside Polymorph.

## BEFORE A GIG

1. Copy the Polymorph folder to the computer; run Polymorph.exe once and let it prepare the engine.
2. Windows: the output at **48000 Hz**; power plan **High performance**; Wi-Fi, Bluetooth and notifications off; close everything else.
3. In Polymorph: choose OUT, the MIDI devices and their roles; load your mapping set.
4. **LOAD PROJECT** your set; check **MONO** once.
5. Run the soak test on that computer, output and power: **PASS**.
6. Charger connected. If the PA hums with the charger in, it is a ground loop: try another socket on the same strip as the mixer, or a DI box with ground lift.
7. Check the clock chain to your other gear.
8. Watch HEALTH in the first minutes: LOST 0, DROPS 0.

# 16 Quick recipes

### Bass pluck

Saw; Sub 0.5; Cutoff 400 Hz; Emphasis 0.3; Contour +0.5, filter Decay 0.15, Sustain 0.1; loudness Decay 0.2, Sustain 0.5, Release 0.15. Drive 40 % for weight.

### Acid line

Saw; Cutoff 380 Hz; Emphasis 0.8; Contour +0.45, filter Decay 0.18, Sustain 0.05; sixteen sixteenths; Accent on steps 3, 6, 9, 12; Slide before the jumps; Accent 80 %, Slide 70 ms.

### Pad that moves

Saw; OSC 2 saw, Fine +7, level 1.0; Cutoff 1500 Hz; Attack 0.4, Release 1.0; Drift 60 %; Rev Snd 0.5, reverb Decay 4 s.

### Pumping pad

Kick on T1 (DRUM), a held chord on T2; Ducked by T1, Depth 100 %, Release 250 ms; T2 DUCK all the way up.

### Techno beat

T1 as DRUM. Kick on 1, 5, 9, 13; open hat on 3, 7, 11, 15; closed hat on the others; snare and clap on 5 and 13. Kick from the table in chapter 6.

### Rumble kick

Four-on-the-floor kick; Rumble Amount 80 %, Decay 1.2 s, Tone 110 Hz, Level 60 %. Longer Decay and lower Tone for a deeper tail.

### Rolling hats

Closed hat on all sixteen; every second one Velocity 70; Kit Modulation `velocity › chh_decay` +0.3.

### Hats that never repeat

DRUM track RANDOM: TUNE, Amount 40 %. Press DICE until you like the run.

### Shuffle

Swing 58–62 %, grid 1/16; leave the kick track's own Swing at 50 % for a straight kick.

### Zap or laser

Pitch Env +24, P.Decay 0.08; short loudness Decay, Sustain 0.

### Thin, nasal lead

Filter BP, Cutoff 1500 Hz, Emphasis 0.5; move Cutoff.

### FM bell

Sine and sine; OSC 2 Octave +1, Semitone +10, level 0; FM 40 %; loudness Decay 1.5 s, Sustain 0, Release 1.5 s; some Rev Snd.

### FM bass

Sine and sine; OSC 2 Octave +1, level 0; FM 50 %; Decay 0.15, Sustain 0.15. Lock FM higher on a few steps.

### Ring clang

Triangle OSC 1, sine OSC 2 at Octave +1, Semitone +6; Ring 100 %; short Decay; echo.

### Bleeps in time

LFO Wave S&H, Sync 1/16, Depth 1; mod matrix `lfo1 › pitch` 0.5; hold a note with the clock playing.

### A melody from the filter

Osc 1 at 0; OSC on; Emphasis 1.00; Key Track 100 %; Cutoff 262 Hz. For zaps: Contour +0.7, filter Decay 0.18, Sustain 0.

### Frozen echo and a dive

Send a phrase to the tape echo; press HOLD; stop playing; pull the fader mapped to Repeat Rate.

### Dub echo on one note

Sends at 0; on one step lock Echo Snd to 1; echo time from division 1/8, Heads 1+2, Intensity 0.6.

### Evolving loop

Track 1: 16 steps. Track 2: length 5. Track 3: length 7 at Speed 1/2x. Chance 50 % on a few steps.

# 17 Genre starters

Eight starting points, one for each corner of the music Polymorph was made for. Each is a full pattern with sounds and effects. Build one, play it, then follow the *Explore* ideas until it is no longer a recipe but your own track.

### HOW TO BUILD A STARTER

1. Pick an empty pattern (1–16) and set the **TEMPO** and **SWING**. On the SEQ page set each track to **SYNTH** or **DRUM**, and its **LENGTH** and **SPEED** if given.
2. **Drums:** select the DRUM track, **REC** on, choose the drum above the steps and click the steps of its row. For a soft or special hit, select the step and set **VELOCITY**, **CHANCE** or **COND** in the step editor.
3. **Notes and chords:** select the SYNTH track, REC off, set the sound. REC on, play the note or chord, then click the steps that should have it. Set **LENGTH**, **ACCENT** and **SLIDE** in the step editor or with the A and S marks.
4. Set the effects on the FX page, press PLAY, and adjust levels in the mixer strip.
5. **SAVE PROJECT** with the style's name. Effects and groove belong to the whole instrument, so a project keeps the starter exactly as it is.

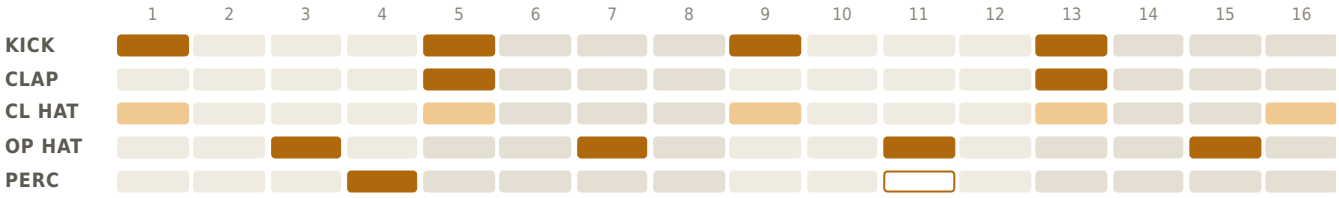
hit · soft hit (lower Velocity) · Accent · Chance or condition

Notes are named as the step editor's **NOTE** knob shows them. The LFO's rate and shape are shared by all tracks: one LFO idea per starter.

The settings are starting points, not rules. Values not named stay as they come.

## Techno

TEMPO 132 · SWING 50 % · T1 DRUM · T2 BASS · T3 STAB




Perc on 11 with Cond 1:2 · closed hats at Velocity 60

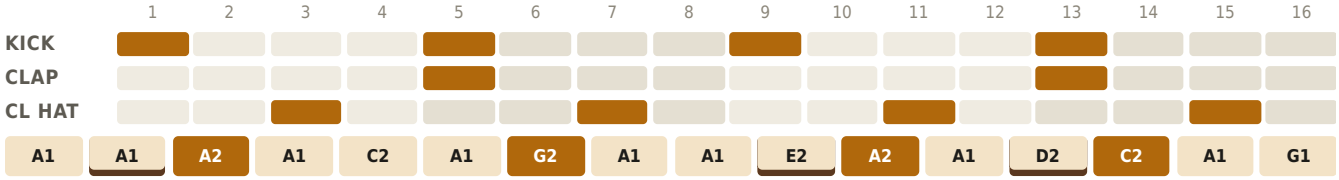
| | |
|---|---|
| **T1 KIT** | Kick: the "punchy techno kick" of chapter 6 (Tune 48 Hz, Pitch Env 70 %, P.Decay 40 ms, Click 40 %, Decay 0.30 s, Drive 40 %, Tone 70 %). Cl hat Decay 40 ms; Op hat Decay 0.25 s; Perc Tune 300 Hz, Decay 0.12 s. |
| **T2 BASS** | Saw; Cutoff 300 Hz, Emphasis 0.3, Contour +0.4, filter Decay 0.12, Sustain 0; loudness Decay 0.12, Sustain 0; Drive 50 %. Note A1 on every step *without* a kick (2, 3, 4, 6, 7, 8 …), Length 0.5. |
| **T3 STAB** | Sine and sine; OSC 2 Octave +1, Osc 2 level 0; FM 35 %; Cutoff 2500 Hz; loudness Decay 0.15, Sustain 0. A minor chord (A3 C4 E4) on step 7, and on 15 with Cond 2:2. Echo Snd 0.4. |
| **EFFECTS** | Rumble Amount 80 %, Decay 1.2 s, Tone 110 Hz, Level 60 %. Ducked by T1, Depth 80 %, Release 150 ms; DUCK up on T2. Echo time from 1/8, Heads 1+3, Intensity 45 %. |

### EXPLORE

- Ride T2's **CUTOFF** between 150 and 900 Hz: the whole track breathes.
- RANDOM on T2: CUTOFF, Amount 30 %. Press **DICE** until the bass line talks.
- Rumble **TONE** down to 60 Hz and Decay to 2 s for a dark warehouse.
- Lock a longer kick Decay (0.8 s) on step 13 only: a boom at the end of every bar.

## *Acid*

TEMPO 130 · SWING 54 % · T1 DRUM · T2 ACID LINE




T2: filled = Accent · underlined = Slide into the next step

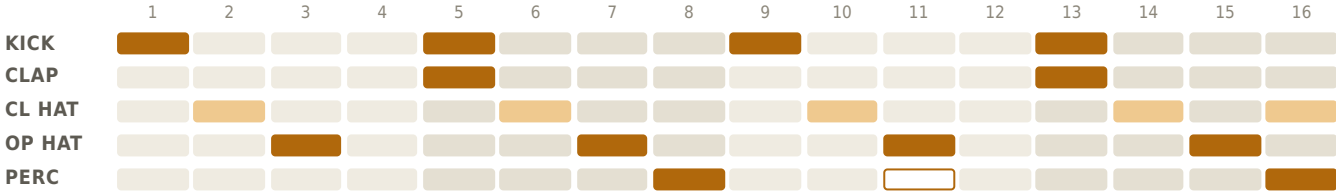
| | |
|---|---|
| **T2 SOUND** | Saw; Cutoff 380 Hz, Emphasis 0.8, Contour +0.45, filter Decay 0.18, Sustain 0.05; loudness Decay 0.25, Sustain 0.6, Release 0.05; Drive 40 %. |
| **GROOVE** | FX page, GROOVE: **ACCENT** 80 %, **SLIDE** 70 ms. |
| **EFFECTS** | Echo Snd 0.25 on T2; echo time from 1/8, Heads 1+2, Intensity 50 %. Reverb low. |

**EXPLORE**

- This is how acid is played: map **CUTOFF**, **EMPHASIS** and **CONTOUR** to three knobs of the controller (MAP mode) and ride them for minutes.
- Push **DRIVE** to 80 % and **EMPHASIS** to 0.95: the line starts to scream.
- Set T2's Length to 12 or 15 against the 16-step kick: the line turns around itself.
- RANDOM on T2: PITCH, Amount 15 %, then DICE: a new line that still repeats.

## *House*

TEMPO 124 · SWING 58 % (GRID 1/16) · T1 DRUM AT SWING 50 % · T2 CHORDS · T3 BASS




Perc on 11 with Chance 50 % · closed hats at Velocity 70




T2: Am7 chord (A3 C4 E4 G4), Length 0.6 · outlined = Cond 1:2




T3: off-beat bass, Length 0.6

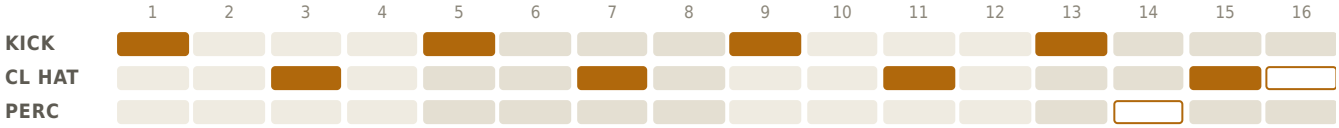
| | |
|---|---|
| **T1 KIT** | Kick: Tune 52 Hz, Pitch Env 50 %, P.Decay 50 ms, Click 20 %, Decay 0.40 s, Drive 10 %, Tone 50 %. Clap Decay 0.3 s, Spread 50 %. Perc as a conga: Tune 400 Hz, Decay 0.08 s. |
| **T2 CHORDS** | Voices 8. Saw; OSC 2 saw, Fine +8, level 0.7; Cutoff 1200 Hz, Emphasis 0.2, Contour +0.3, filter Decay 0.2; loudness Attack 0.005, Decay 0.25, Sustain 0, Release 0.2. Rev Snd 0.3, Echo Snd 0.2. |
| **T3 BASS** | Sine; Sub 0.3; Cutoff 600 Hz; loudness Decay 0.2, Sustain 0.4, Release 0.08. |
| **EFFECTS** | Ducked by T1, Depth 50 %, Release 200 ms; DUCK up on T2 and T3. Reverb Decay 1.5 s. Echo time from 1/8, Heads 1, Intensity 30 %. |

**EXPLORE**

- Global **SWING** from 54 to 64 %: from straight to deep shuffle. The kick stays straight.
- A slow sweep: LFO SINE, Rate 0.1 Hz, Depth 1 on T2; mod matrix `lfo1 › cutoff` 0.2.
- Change the chord on loop 2 only: a second chord (Dm7) on step 3 with Cond 2:2, and Cond 1:2 on the Am7.
- Velocity of the claps 90 on 5 and 110 on 13: the second half of the bar leans forward.

## *Dub techno*

TEMPO 120 · SWING 52 % · T1 DRUM · T2 CHORD STAB




Hat 16 at Velocity 50 with Chance 50 % · perc 14 with Cond 1:4




T2: Cm7 (C3 D#3 G3 A#3) · outlined = Cond 2:2

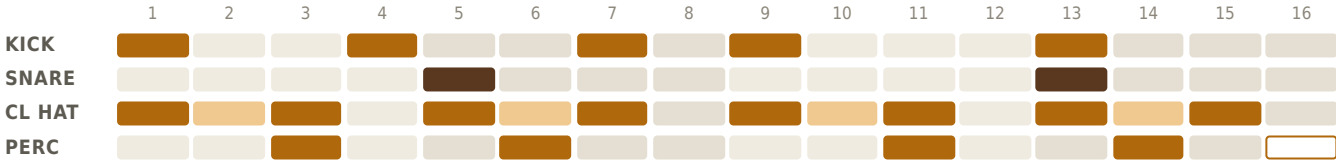
| | |
|---|---|
| **T1 KIT** | Kick: Tune 45 Hz, Decay 0.5 s, Drive 0 %, Tone 40 %. Cl hat Decay 30 ms, Rev Send 0.3. Perc Tune 500 Hz, Decay 0.05 s, Echo Send 0.6. |
| **T2 STAB** | Voices 8. Saw; OSC 2 saw, Fine +6, level 0.8; Cutoff 700 Hz, Emphasis 0.35, Contour +0.25, filter Decay 0.25; loudness Decay 0.3, Sustain 0, Release 0.4. Echo Snd 0.8, Rev Snd 0.4. |
| **MODULATION** | LFO SINE, Sync 1/1, Depth 0.6 on T2; mod matrix `lfo1 › cutoff` 0.3: the stab opens and closes once a bar. |
| **EFFECTS** | Echo time from 1/8, Heads 1+3, Intensity 65 %, Wow 0.3, Level 0.7, To Rev 0.5. Reverb Pre-dly 0.03 s, Decay 6 s, Damping 0.6. Rumble Amount 60 %, Decay 2.5 s, Tone 80 Hz. |

**EXPLORE**

- At the end of a phrase press **HOLD** on the echo, mute T2, let the loop turn, then let go.
- Echo **INTENSITY** briefly over 85 %: the chords swell into a roaring wall. Bring it back down; keep Master in mind.
- Chance 60 % on the stab steps: the chords come and go like weather.
- Ride T2's Cutoff between 400 and 1500 Hz, very slowly. Dub techno is patience.

## *Industrial and noise*

TEMPO 140 · SWING 50 % · T1 DRUM · T2 RING CLANG · T3 SCREAMING FILTER (SPEED 2X)




Snares Accent · perc 16 with Chance 50 % · RANDOM on T1: TUNE, Amount 50 %




T2: outlined = Chance 50 % · RANDOM PITCH 60 %

| | |
|---|---|
| **T1 KIT** | Kick: the "hard, driven kick" of chapter 6 (Tune 55 Hz, Pitch Env 85 %, P.Decay 25 ms, Click 70 %, Decay 0.22 s, Drive 80 %, Tone 85 %). Snare Tune 150 Hz, Tone 20 %, Snappy 90 %, Decay 0.3 s. Perc as metal: Tune 700 Hz, Pitch Env 60 %, Decay 0.06 s. Kit Filter Drive 80 %, Emphasis 0.4. |
| **T2 CLANG** | Triangle OSC 1, sine OSC 2 at Octave +1, Semitone +6; Ring 100 %; Drive 70 %; loudness Decay 0.12, Sustain 0. Echo Snd 0.5. |
| **T3 SCREAM** | Osc 1 at 0, Noise 0.2; OSC on; Emphasis 1.00; Key Track 100 %; Cutoff 262 Hz; Contour +0.7, filter Decay 0.18, Sustain 0. Notes C3 on steps 2 and 10. |
| **EFFECTS** | Echo time from 1/16, Heads ALL, Intensity 80 %, Wow 0.6. Reverb Decay 2 s, Damping 0.2. |

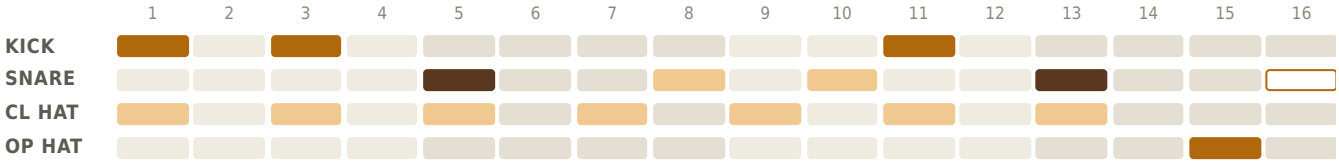
**EXPLORE**

- Echo **INTENSITY** over 85 %: feedback builds by itself. Then **HOLD**, and the noise becomes a loop.
- Kit Filter Drive 100 % and Cutoff down to 800 Hz: the whole kit through a broken speaker.
- **DICE** on every phrase: the metal never hits the same way twice.
- Lock T3's Cutoff to a different note on step 10: a two-note siren.

## *Jungle*

TEMPO 170 · SWING 50 % · T1 DRUM · T2 BASS (SPEED 1/2) · T3 PAD (SPEED 1/8)

Polymorph has no samples, so the break is built from the kit: ghost notes, velocity and microtiming do the work.




Snares 5 and 13 Accent · ghosts 8, 10 at Velocity 45, Micro +3 · 16 at Velocity 40, Chance 60 % · hats Velocity 70

| E1 | 2 | 3 | 4 | 5 | 6 | E1 | G1 | 9 | 10 | 11 | 12 | D1 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

T2 (Speed 1/2, so 16 steps last two bars): Lengths 4, 1, 4, 4 · underlined = Slide into the G

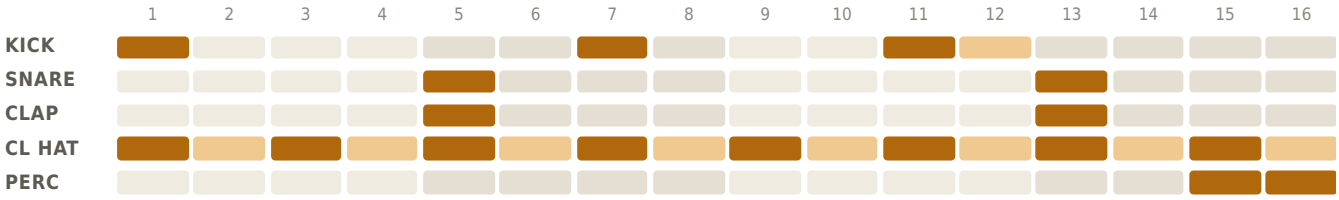
| | |
|---|---|
| **T1 KIT** | Kick: Tune 55 Hz, Pitch Env 60 %, P.Decay 30 ms, Click 50 %, Decay 0.25 s, Tone 80 %. Snare: Tune 200 Hz, Tone 40 %, Snappy 75 %, Decay 0.18 s. Cl hat Decay 30 ms; Op hat Decay 0.15 s. Kit Filter Drive 40 % for grit. |
| **T2 BASS** | A moving bass from two saws: Saw; OSC 2 saw, Fine +12, level 1.0; Drift 50 %; Drive 50 %; Cutoff 350 Hz, Emphasis 0.25; loudness Sustain 1, Release 0.2. LFO TRIANGLE, Sync 1/2, Depth 0.5; mod matrix `lfo1 › cutoff` 0.25. |
| **T3 PAD** | Track Length 4 at Speed 1/8 (two bars). One chord on step 1 (E3 G3 B3 D4), Length 4.00. Saw; OSC 2 saw, Fine +7; Cutoff 1800 Hz; loudness Attack 0.8, Release 2.5; Rev Snd 0.6. |
| **EFFECTS** | Reverb Decay 3 s. Echo time from 1/8, Heads 1+2, Intensity 40 %. |

**EXPLORE**

- RANDOM on T1: VELOCITY, Amount 40 %: the break sounds played, not programmed.
- Snare roll on 13–16 with Cond FILL; hold **FILL** (key V) before the drop.
- A second DRUM track of hats only, at Speed 2x with Chance 70 %: rattling top end.
- Tape stop: **HOLD** the echo with Time from Repeat Rate and pull the mapped Repeat Rate fader down.

## *Electro*

TEMPO 128 · SWING 50 % · T1 DRUM · T2 FM BASS · T3 BLEEPS




Kick 12 at Velocity 70 · hats alternate Velocity 100 and 60 · Perc 16 locked to Tune 260 Hz

| E1 | 2 | 3 | E1 | 5 | 6 | G1 | 8 | E1 | 10 | E2 | D2 | 13 | E1 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

T2: Length 0.5 · filled = FM locked to 80 %

| | |
|---|---|
| **T1 KIT** | Kick: Tune 50 Hz, Pitch Env 80 %, P.Decay 60 ms, Click 30 %, Decay 0.6 s, Tone 60 %. Snare Tune 180 Hz, Snappy 60 %, Decay 0.15 s. Cl hat Decay 25 ms. Perc Tune 180 Hz, Pitch Env 40 %, Decay 0.2 s. |
| **T2 FM BASS** | Sine and sine; OSC 2 Octave +1, level 0; FM 50 %; loudness Decay 0.15, Sustain 0.15. Lock FM 80 % on step 11. |
| **T3 BLEEPS** | Square; Cutoff 3000 Hz; loudness Decay 0.08, Sustain 0. LFO S&H, Sync 1/16, Depth 1; mod matrix `lfo1 › pitch` 0.5. One note (A4) on steps 3, 6, 10, 14; Echo Snd 0.5. |
| **EFFECTS** | Echo time from 1/16, Heads 1+2, Intensity 40 %. Reverb short: Decay 0.8 s. |

**EXPLORE**

- **DICE** does not touch the S&H line; change the LFO **SYNC** to 1/8 or 1/32 for other bleeps.
- Ring 60 % on the bass: colder, more metallic.
- Zaps: a fourth track with Pitch Env +24, P.Decay 0.08, short Decay, on step 16 with Cond 1:4.
- Hats on a 15-step track: the accents walk across the bar.

## *Ambient*

TEMPO 70 · T1 PAD (SPEED 1/8) · T2 BELLS (LENGTH 7, SPEED 1/2) · T3 WIND (SPEED 1/8) · NO DRUMS

| Dm7 | 2 | 3 | 4 | A#maj7 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

T1, Length 8 at Speed 1/8 (four bars): Dm7 (D3 F3 A3 C4), A#maj7 (A#2 D3 F3 A3), each Length 4.00

| A4 | 2 | E5 | 4 | 5 | D5 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

T2, Length 7: E5 with Chance 50 % · D5 with Cond 1:2

| | |
|---|---|
| **T1 PAD** | Voices 8. Saw; OSC 2 triangle, Octave +1, Fine +7, level 0.6; Cutoff 900 Hz, Emphasis 0.2, Contour +0.3, filter Attack 1.2 s; loudness Attack 1.5 s, Decay 2 s, Sustain 0.8, Release 3 s; Drift 70 %; Rev Snd 0.8, Echo Snd 0.3. |
| **T2 BELLS** | The FM bell of chapter 16: sine and sine, OSC 2 Octave +1, Semitone +10, level 0; FM 40 %; loudness Decay 1.5 s, Sustain 0, Release 1.5 s. Echo Snd 0.5, Rev Snd 0.6. |
| **T3 WIND** | Osc 1 at 0, Noise 0.5; Filter BP, Cutoff 800 Hz, Emphasis 0.7; loudness Attack 2 s, Release 3 s. Track Length 2, one note on step 1, Length 2.00. LFO RANDOM, FREE, Rate 0.1 Hz, Depth 1; mod matrix `lfo1 › cutoff` 0.4. |
| **EFFECTS** | Echo Time from Repeat Rate 0.55 s, Heads 2+3, Intensity 55 %, Wow 0.5, To Rev 0.6. Reverb Pre-dly 0.05 s, Decay 9 s, Damping 0.4, Level 0.7. |

**EXPLORE**

- RANDOM on T2: PITCH, Amount 30 %: the bells wander off the chord. DICE until the line is beautiful.
- Drift 100 % on the pad: the chord becomes a choir that cannot agree.
- T2 Length 5 against the pad's four bars: the melody never lands twice in the same place.
- Echo **HOLD** and let the tape loop carry the piece while you change the pad.

# 18 If something goes wrong

| WHAT HAPPENS | WHAT TO CHECK |
|---|---|
| **No sound** | AUDIO ON lit? The right OUT? Master up? Polymorph the active window (computer keys)? |
| **My output is faint in the OUT list, or missing** | It is not at 48 kHz: set it in Windows sound settings, reopen Polymorph. Connect devices before opening. |
| **The pattern plays, my playing is silent** | LIVE M lit? LIVE Level down? A DRUM track selected (only drum notes sound)? |
| **Sequencer silent** | PLAY pressed? Steps lit? Track length covers them? Track muted or Level down? A Cond or Chance stopping them (read the amber line)? |
| **Clicking a step does nothing** | REC is off (`REC IS OFF`). Press REC. |
| **A knob changes the wrong thing** | Which track is selected? Is a step held (`STEP n HELD`)? |
| **Recording does nothing** | REC lit and clock playing, or STEP REC lit and clock stopped? The right track selected? |
| **A track does not come back after unmuting** | It waits for its step 1; a long or slow track can take several bars. |
| **The pattern did not change when clicked** | It changes at the end of the bar. |
| **No echo or reverb** | FX ON lit? A send above 0? Echo / Reverb Level above 0? |
| **Echo on my playing but not on the sequence** | LIVE Echo / Reverb are for your playing only. Raise the track's Echo Snd / Rev Snd. |
| **The echo never stops** | HOLD lit, or Intensity above self-oscillation. Lower it, or switch FX ON off to clear the tape. |
| **A DRUM track or drum is silent** | Drum M lit? Drum Level? Kit Filter Cutoff down? |
| **Slide does nothing** | The next step needs a note; SYNTH tracks only. |
| **No ducking** | Ducked by set to the kick's track? DUCK up on the track that should dip? Depth up? |
| **No rumble** | Rumble Amount above 0? Its drum playing on a DRUM track? Level up? |
| **The filter does not sing** | OSC lit? Emphasis above about 87 %? A note playing? |
| **FM or Ring do nothing** | They depend on OSC 2's tuning; its level may be 0. |
| **RANDOM does nothing** | Amount up and something chosen? It acts on sequenced steps only. |
| **The arpeggiator ignores Rate** | The clock is playing: Division is in use. |
| **The LFO's Rate does nothing** | Sync is on a division. Choose FREE. |
| **High notes are dull or quiet** | Raise Key Track or Cutoff. |
| **The pads answer late** | SETUP: WASAPI, Block 512, Card buffer LOW (audio off to change). |
| **Crackles, or HEALTH shows LOST / DROPS** | Block 1024, Card buffer HIGH, or LIGHT LAPTOP; close other programs; run the soak test. |
| **MIDI device missing** | REFRESH (FX page, MIDI). Ports open on AUDIO ON. |
| **The fader box plays notes / the pads do not play** | Wrong role: CTRL for faders and knobs, NOTES or BOTH for pads and keys. |
| **A mapped knob jumps or covers part of its travel** | MAP mode: check POSITIONS / STEPS (press R to change); TAKEOVER PICK-UP against first-touch jumps; check its MIN / MAX. |
| **A mapped knob does nothing** | TAKEOVER PICK-UP waiting (the display says which way); or the device's role is NOTES. |

| | |
|---|---|
| **A pad no longer plays a note** | It was linked to a button: MAP, click that button, Delete. |
| **Controls do nothing and have amber frames** | MAP mode is on: press MAP again. |
| **Notes on MIDI channel 16 are silent** | Channel 16 is the control channel. |
| **Other gear does not follow the tempo** | OUT chosen and SEND CLOCK lit? Audio on? The other device set to follow external clock? |
| **Everything sounds narrow** | MONO is lit. |
| **LOAD did nothing** | Read the display: wrong kind of file, or waiting for the bar line. |
| **Polymorph says it did not close properly** | Answer Yes to restore the last autosave. |
| **Stuck notes or a runaway echo** | PANIC (`Esc`). STOP releases sequenced notes. |
| **I cannot leave full screen** | `F11` (`Esc` is PANIC). |

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